

Categories, Proofs, and Processes:

Lecture 20 Monads

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9 Monads

9.1 Monoids on the Category of Endofunctors

For $[\mathcal{C}, \mathcal{C}]$ the category of endofunctors, with

- objects - functors

$$F : \mathcal{C} \times \mathcal{C}$$

- morphisms - natural transformations

$$\phi : F \rightarrow G$$

Define

$$F \otimes G := FG$$

$$I := 1_{\mathcal{C}}$$

Note 9.1. In what sense can I have a natural transformation in parallel?

Think of G as some structure on the inside, and F as structure on the outside. Do some operation on G , also doing the same operation on F in parallel

9.2 Natural transformations "in parallel"

Example 9.2. Define

$$\mathbf{List} : \mathbf{Set} \rightarrow \mathbf{Set}$$

$$X \mapsto \mathbf{List} X$$

where $\mathbf{List}(X)$ is the set of finite lists and elements of X

$$f : X \rightarrow Y$$

$$\mathbf{List}(f) : \mathbf{List} X \rightarrow \mathbf{List} Y$$

$$[x_1, \dots, x_k] \mapsto [f(x_1), \dots, f(x_k)]$$

Define also

$$\mathbf{BTree} : \mathbf{Set} \rightarrow \mathbf{Set}$$

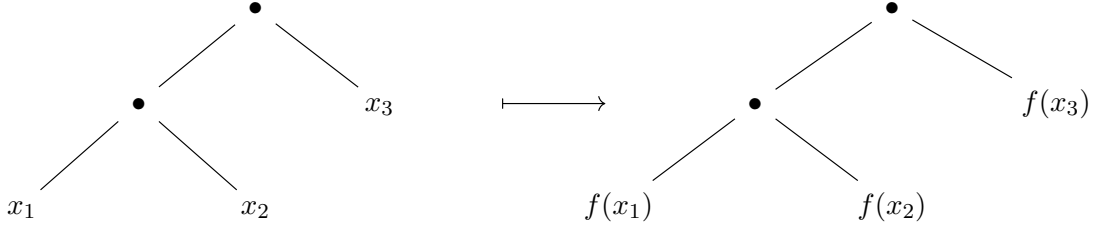
$$X \mapsto \mathbf{BTree} X$$

where $\mathbf{BTree}(X)$ is the set of finite binary trees with leaves labelled by elements of X

$$\left\{ \begin{array}{c} \bullet \\ \swarrow \quad \searrow \\ \bullet \quad \quad x_3, \dots \\ \swarrow \quad \searrow \\ x_1 \quad x_2 \end{array} \right\}$$

$$f : X \rightarrow Y$$

$$\mathbf{BTree}(f) : \mathbf{BTree} X \rightarrow \mathbf{BTree} Y$$



Theorem 9.3 (Informal). *It is possible to apply natural transformations "over" or "under" functors.*

Let

$$\phi : F \rightarrow F'$$

Definition 9.4 ("Over").

$$\phi \otimes G : FG \rightarrow F'G$$

$$(\phi \otimes G)_A := \phi_{G A} : FGA \rightarrow FG' A$$

which is natural as it follows from the naturality of ϕ

Definition 9.5 ("Under").

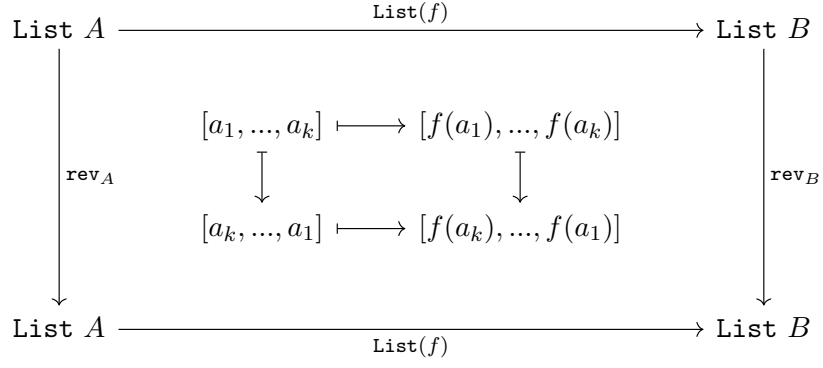
$$G \otimes \phi : GF \rightarrow GF'$$

$$(G \otimes \phi)_A := G(\phi_A) : GFA \rightarrow GF' A$$

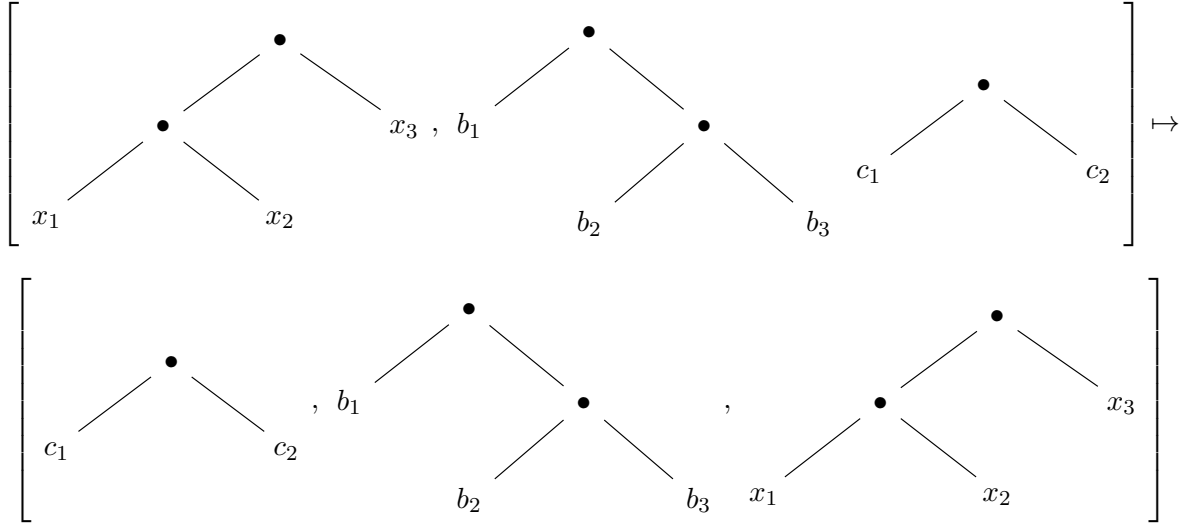
Example 9.6.

$$\mathbf{rev} : \mathbf{List} \rightarrow \mathbf{List}$$

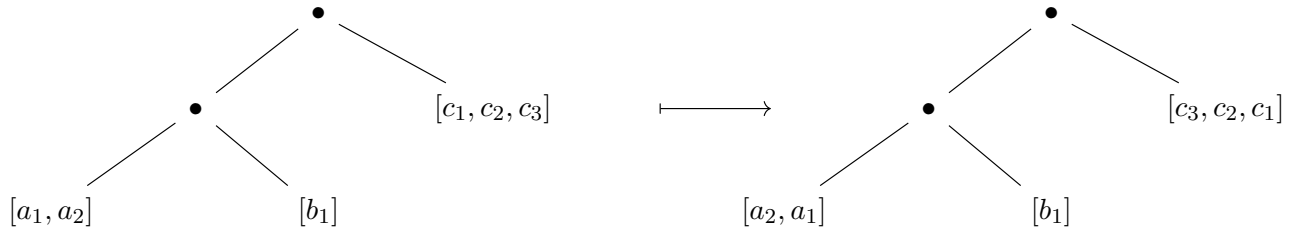
$$\mathbf{rev}_A([a_1, \dots, a_k]) = [a_k, \dots, a_1]$$



$$\text{rev} \otimes \text{BTree} : \text{List} \circ \text{BTree} \rightarrow \text{List} \circ \text{BTree}$$



$$\text{BTree} \otimes \text{rev} : \text{BTree} \circ \text{List} \rightarrow \text{BTree} \circ \text{List}$$



9.3 Results

Proposition 9.7. *For natural transformations*

$$\phi : F \rightarrow F', \quad \psi : G \rightarrow G'$$

we have

$$(\phi \otimes G') \circ (F \otimes \psi) = (F' \otimes \psi) \circ (\phi \otimes G) =: \phi \otimes \psi$$

Proof.

$$(\phi \otimes G')_A \circ (F \otimes \psi)_A = \phi_{G' \ A} \circ F(\psi_A)$$

$$(F' \otimes \psi)_A \circ (\phi \otimes G)_A = F'(\psi_A) \circ \phi_{G \ A}$$

Naturality of ϕ :

$$\begin{array}{ccc} FGA & \xrightarrow{F(\psi_A)} & FG'A \\ \phi_{G \ A} \downarrow & & \downarrow \phi_{G' \ A} \\ F'GA & \xrightarrow{F'(\psi_A)} & F'G'A \end{array}$$

□

Theorem 9.8. $[\mathcal{C}, \mathcal{C}]$ *is a strict monoidal category.*

Example 9.9. The monoids in $[\mathcal{C}, \mathcal{C}]$ are functors F with natural transformations:

$$\mu : FF \rightarrow F$$

$$\eta : 1_{\mathcal{C}} \rightarrow F$$

that are associative and unital.

Associativity:

$$\begin{array}{ccc} FFF & \xrightarrow{\mu \otimes F} & FF \\ F \otimes \mu \downarrow & & \downarrow \mu \\ FF & \xrightarrow{\mu} & F \end{array}$$

Left Unit:

$$\begin{array}{ccc} F = 1_{\mathcal{C}} \circ F & \xrightarrow{\eta \otimes F} & FF \\ & \searrow 1_F & \downarrow \mu \\ & & F \end{array}$$

Right unit:

$$\begin{array}{ccc} F = F \circ 1_{\mathcal{C}} & \xrightarrow{F \otimes \eta} & FF \\ & \searrow 1_F & \downarrow \mu \\ & & F \end{array}$$

9.4 Monads

Definition 9.10. A monad is a monoid in $[\mathcal{C}, \mathcal{C}]$.

Note 9.11. Think of monads as some sort of inductive instruction. E.g. There's always a natural way of flattening lists of lists, or gluing trees of trees.

Example 9.12. `List` is a monad. $\mu_A : \text{List}^2 A \rightarrow \text{List } A$ is "flatten".

$$[[a_1^{(1)}, \dots, a_{k_1}^{(1)}], [a_1^{(n)}, \dots, a_{k_n}^{(n)}]] \mapsto [a_1^{(1)}, \dots, a_{k_1}^{(1)}, \dots, a_1^{(n)}, \dots, a_{k_n}^{(n)}]$$

$$\eta_A : A \rightarrow \text{List } A$$

$$a \in A \mapsto [a]$$

Associativity:

$$\begin{array}{ccc} \text{List}^3 A & \xrightarrow{\text{List} \otimes \mu} & \text{List}^2 A \\ \mu \otimes \text{List} \downarrow & & \downarrow \mu \\ \text{List}^2 A & \xrightarrow{\mu} & \text{List } A \end{array}$$

Unit:

$$\begin{array}{ccc} [a, b, c] & \xrightarrow{\quad\quad\quad} & [[a], [b], [c]] \\ \downarrow & & \downarrow \\ \begin{array}{ccc} \text{List } A & \xrightarrow{\text{List} \otimes \eta} & \text{List}^2 A \\ \eta \otimes \text{List} \downarrow & \searrow 1_A & \downarrow \mu \\ \text{List}^2 A & \xrightarrow{\mu} & \text{List } A \end{array} & & \\ [[a, b, c]] & \xrightarrow{\quad\quad\quad} & [a, b, c] \end{array}$$